Industrial University of Ho Chi Minh City

CQRS Pattern

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CQRS là gì?

CQRS (Command Query Responsibility Segregation) là một mẫu kiến trúc phần mềm tách biệt rõ ràng giữa:

Solution

- Command (Ghi) → thay đổi trạng thái hệ thống
- Query (Đọc) → chỉ lấy dữ liệu, không thay đổi gì

Nguyên lý cốt lõi:

“Không nên dùng cùng một model cho cả đọc và ghi”

Example: Hệ thống bán vé tàu

- **Command:**
  - Đặt vé
  - Hủy vé
  - Thanh toán
- **Query:**
  - Xem danh sách vé
  - Xem lịch sử đặt vé
  - Tìm chuyến tàu

CQRS tách riêng:

- Service xử lý đặt vé ≠ Service hiển thị vé

Kiến trúc tổng thể

1. Command Side (Write)

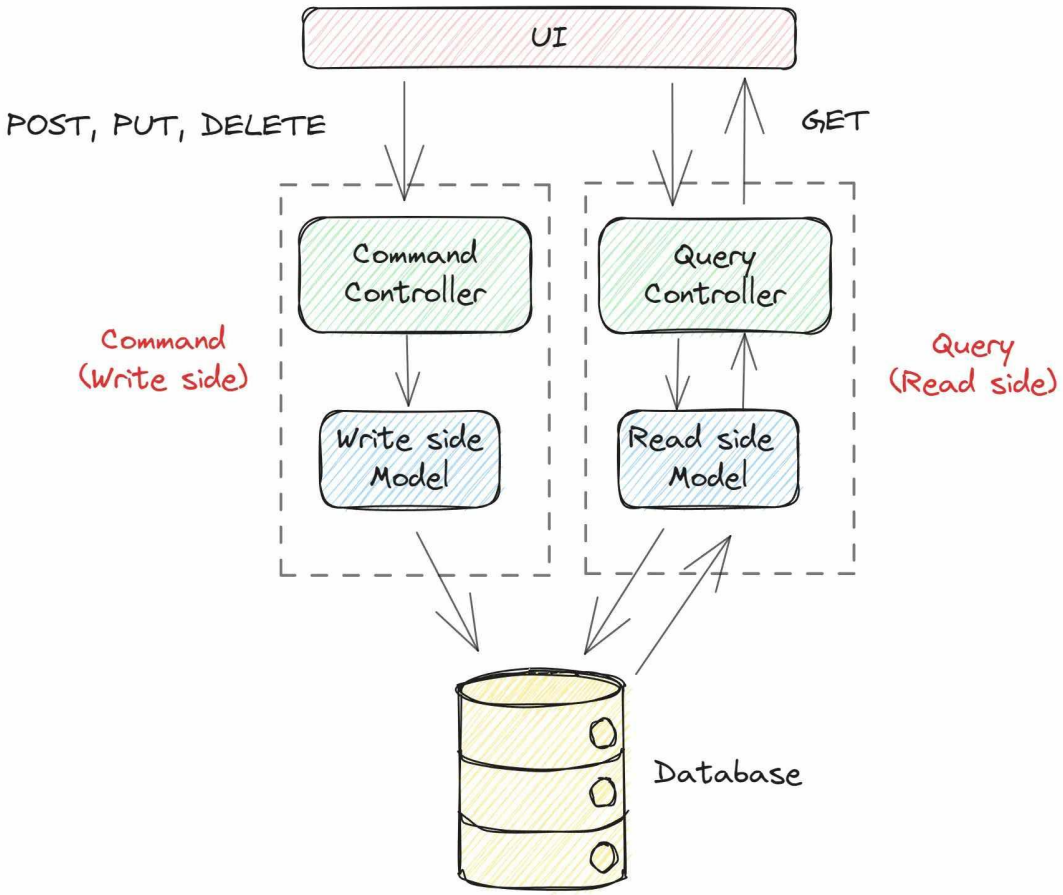
- Nhận request thay đổi dữ liệu
- Validate logic nghiệp vụ
- Ghi vào DB (Write DB)

2. Query Side (Read)

- Chỉ đọc dữ liệu
- Dùng Read DB (được tối ưu riêng)

3. Event / Sync (optional)

- Đồng bộ dữ liệu giữa Write → Read
- Thường dùng:
  - Event Bus (Kafka, RabbitMQ)
  - Event Sourcing



So sánh CRUD vs CQRS

Tiêu chí	CRUD	CQRS
Model	1 model	2 model
Performance	Trung bình	Tối ưu cao
Scale	Khó scale	Scale tốt
Complexity	Thấp	Cao
Phù hợp	App nhỏ	System lớn

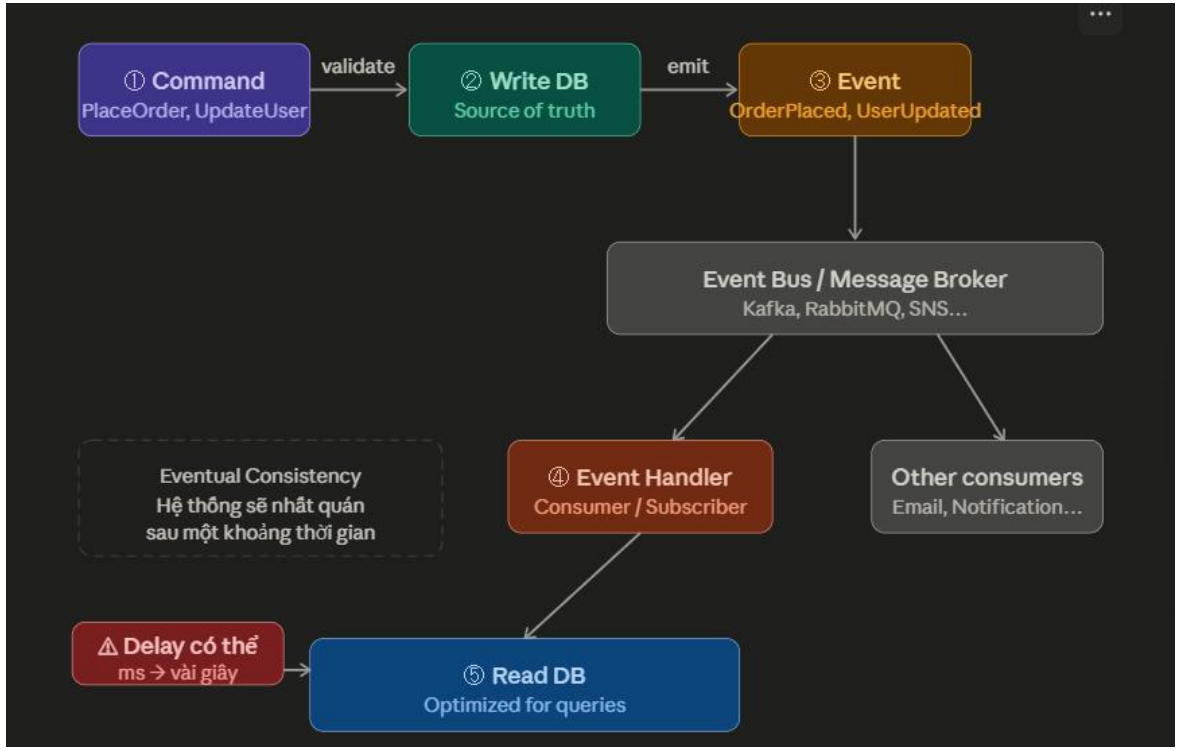
Event-driven + CQRS

Event là gì?

- Flow: Command → Write DB
- Emit Event
- Update Read DB

Giải thích:

- Eventual consistency là gì
- Tại sao Read DB có thể bị delay



Scaling trong CQRS

Scale Read:

- nhiều replica

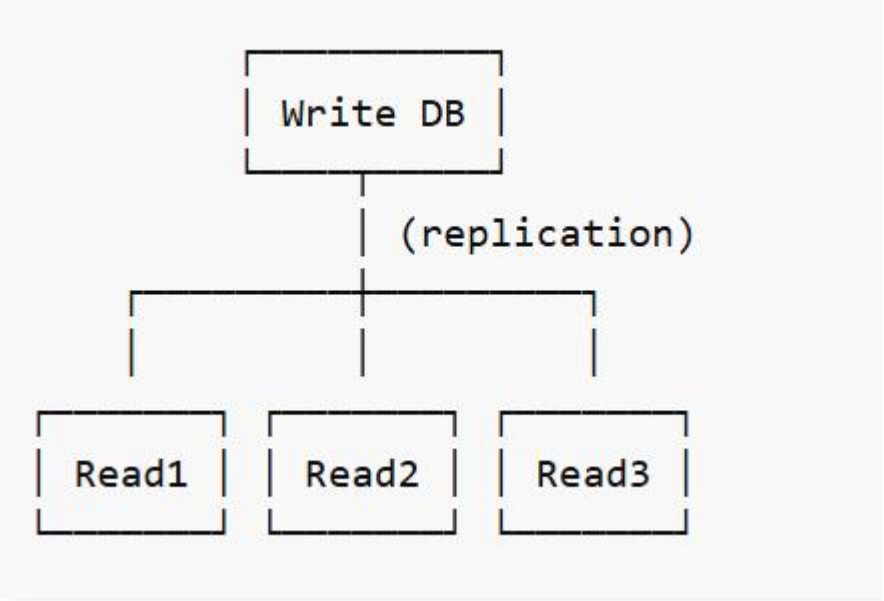
Scale Write:

- shard

Case:

Facebook:

- write ít
- read cực nhiều



CQRS + Microservices

CQRS thường đi với:

- Event-driven architecture
- Microservices

Pattern-mỗi service có:

- DB riêng
- read model riêng

PRACTICE & DESIGN

- Đề bài: Hệ thống bán vé tàu
- Yêu cầu: Thiết kế CQRS cho
  - Đặt vé
  - Hủy vé
  - Xem vé
  - Tìm chuyến

Recommendation

- Command:
  - BookTicket
  - CancelTicket
- Query:
  - GetTickets
  - SearchTrips
- Event:
  - TicketBooked
  - TicketCanceled

Question

- Nếu Read DB bị lag thì sao?
- Có cần transaction không?
- Khi nào không dùng CQRS?

Summary

- CQRS = tách Read / Write
- Tăng performance + scale

Bài 1: Todo App với CQRS

Yêu cầu:

- Xây dựng API:
- **Command:**
  - POST /todos → tạo todo
  - PUT /todos/:id → update
  - DELETE /todos/:id
- **Query:**
  - GET /todos → lấy danh sách
  - GET /todos/:id → chi tiết

Bài 2: Order System (có Event)

Yêu cầu:

- **Command:**
  - Tạo đơn hàng
  - Hủy đơn
- **Query:**
  - Xem danh sách đơn hàng

Bài 3: Train Ticket System

Yêu cầu:

- **Command:**
  - Book Ticket
  - Cancel Ticket
- **Query:**
  - Get Tickets
  - Search Trips

Bài 4: CQRS + SCALING + MICROSERVICE

Tách thành 2 service

Yêu cầu:

- Tách thành:
  - order-command-service
  - order-query-service

Giao tiếp:

  - Kafka / RabbitMQ (giả lập bằng eventBus)

Q&A

Thank you very much!

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Event Sourcing

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Event Sourcing Architecture

- What is Event Sourcing?
- Why use it?
- How it works?

- Keywords: Event, Replay, State

Problem of CRUD

- Only stores final state
- No history
- Hard to debug

- Example: UPDATE account SET balance = 1000

Core Idea

- Do not store state
- Store events

- State = replay events

How It Works

- Command → Event → Store → Projection → UI

Architecture Components

- Command Handler
- Event Store
- Event Bus
- Projection

Replay State

- State is rebuilt by replaying events

Snapshot Optimization

- Use snapshot to avoid replaying all events

CRUD vs Event Sourcing

- CRUD: State
- Event Sourcing: Events
- Better audit and debugging

When to Use

- Banking, E-commerce, Audit systems
- Not for simple CRUD apps

Question

- Large scale but slow?
- → Use snapshot + CQRS + async projection

Event Sourcing - Exercises

- 5 practical exercises from basic to advanced

Exercise 1: Bank Account

- Events: AccountCreated, MoneyDeposited, MoneyWithdrawn
- Store events only (no direct state)
- Implement apply(event), replay(events)
- Output: balance computed from events

Exercise 2: Time Travel / Undo

- Implement undoLastEvent() or getStateAt(index)
- Allow viewing past states
- Hint: replay until index

Exercise 3: Projection (Read Model)

- Separate Write & Read model
- Build accountSummary from events
- Fields: balance, totalDeposits

Exercise 4: Snapshot Optimization

- Create snapshot every N events
- Store snapshot + new events
- Replay from snapshot instead of beginning

Exercise 5: Order System + CQRS

- Events: OrderCreated, ItemAdded, ItemRemoved, OrderConfirmed
- Write side: command -> event
- Read side: projection (totalPrice, status)
- Apply CQRS separation

Question

- What if replay on every query?
- Wrong: replay all events each time
- Correct: use projection + CQRS + async

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Saga Pattern

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Saga Pattern - 3 Hour Lecture

- Distributed transactions in Microservices
- Case Study: Shopee Order Flow

Problem: Distributed Transaction

- Multiple services
- No global ACID
- Need consistency

Why Not 2PC?

- Slow
- Locks resources
- Hard to scale

Saga Pattern Concept

- Split into local transactions
- Each step emits event
- Failure → compensation

Core Components

- Local Transaction
- Event / Message
- Compensation Logic

Saga Flow Example

- Create Order
- Process Payment
- Update Inventory
- Ship Order

Failure Handling

- If step fails → trigger rollback
- Compensation per service

Choreography

- Event-driven
- No central control
- Hard to debug

Orchestration

- Central orchestrator
- Controls flow
- Easier to manage

Compare 2PC vs Saga

- 2PC: Strong consistency
- Saga: Eventual consistency

Shopee Case Study - Overview

- Order Service
- Payment Service
- Inventory Service
- Shipping Service

Shopee Flow

- User places order
- Reserve stock
- Process payment
- Confirm order

Shopee Failure Case

- Payment fails
- Release stock
- Cancel order

Shopee Architecture

- Event Bus (Kafka)
- Microservices
- Async communication

Design Challenges

- Idempotency
- Retry
- Duplicate events

Best Practices

- Use message broker
- Design compensation carefully
- Logging & monitoring

Trap Questions

- Scale but still fail?
- Missing compensation?
- Orchestrator bottleneck?

Summary

- Saga = distributed transaction solution
- Key: events + compensation

Saga Pattern - Practice Exercises

- 6 exercises from basic to advanced
- Focus: Microservices & Distributed Transactions

Exercise 1: Basic Saga (Orchestration)

- Flow: CreateOrder → Payment → Inventory
- Build Saga Orchestrator
- If Payment fails → Cancel Order

Exercise 2: Full Compensation

- Each step has rollback
- Payment → Refund
- Inventory → Restock
- Test failure scenarios

Exercise 3: Choreography

- Event-driven approach
- OrderCreated → PaymentProcessed → InventoryUpdated
- No central orchestrator

Exercise 4: Idempotency

- Handle duplicate events
- Use eventId
- Ignore already processed events

Exercise 5: Retry & DLQ

- Retry failed operations (max 3)
- Push failed events to Dead Letter Queue
- Simulate temporary failures

Exercise 6: Shopee Case Study

- Services: Order, Payment, Inventory, Shipping
- Implement full Saga flow
- Handle failures & compensation
- Add logging & idempotency

Trap Question

- Duplicate payment issue
- Order processed twice
- Find root cause & fix